

Carlos Pineda-Antunez

Curriculum Vitae

Updated September 2026

Carlos Pineda-Antunez

PhD student in Health Economics and Outcomes Research
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Research Profile

Health economist and decision scientist focused on health economics, outcomes research, decision modeling, and public health policy. My work evaluates prevention interventions, health system performance, and the value of health technologies, with applications in colorectal cancer, HIV, maternal health, mental health, and financial protection. I develop decision models, simulation tools, economic evaluations, and reproducible software workflows to support evidence-informed decision making.

Education

- Ph.D in Health Economics & Outcomes Research, University of Washington, 2027
- M.S. in Health Economics, National Institutes of Public Health, 2015
- B.S. in Biomedical Engineering , National Polytechnic Institute, 2011

Professional Experience

2025 - current: Stanford University, California, US Decision Modeler, Department of Health Policy, School of Medicine.

2025-2026: University of Washington, Washington, US Teaching assistant at the Data Science in Health and Outcomes Research Certificate Program, The Comparative Health Outcomes, Policy, and Economics Institute, School of Pharmacy.

2024-2025: University of Minnesota, Minneapolis, US Health Economist, Division of Health Policy and Management, School of Public Health.

2024-2025: University of Washington, Washington, US Health Economist, The Comparative Health Outcomes, Policy, and Economics Institute, School of Pharmacy.

2024 (June - September): Stanford University, California, US Decision Modeler, Department of Health Policy, School of Medicine.

2022 – 2024: Health Research Consortium (CISIDAT), Morelos, Mexico Decision scientist, Division of quantitative methods

2021 – 2022: Center for Research and Teaching in Economics (CIDE) Mexico City, Mexico Decision Modeler, Division of Public Administration

2015 – 2021: National Institute of Public Health (INSP), Morelos, Mexico Health economist, Division of Health Economics and Health Systems Innovations

Software Development and Published Tools

BayCANN for CISNET colorectal cancer models

Code to perform emulator-based Bayesian calibration for the three Colorectal Cancer CISNET models. Publication: [Medical Decision Making](#). Repository: [NCI-CISNET-Colorectal/baycann_cisnet_crc](#).

Cancer modeling with discrete-event simulation

Tutorial and open-source workflow for cancer modeling with discrete-event simulation. Publication: [PharmacoEconomics](#). Repository: [sjpi22/tutorial_cancer_modeling_des](#).

ggpop

R package built on top of ggplot2 to simplify the creation of icon-based population charts. CRAN page: [ggpop](#). Project website: [jurjoraa.github.io/ggpop](#).

Publications

- Servan-Mori E, Cerecero-Garcia D, **Pineda-Antunez C**, Jan S, Hone T, Flamand L, Mohar-Betancourt A, Mendoza MA, Mendez J, Garcia-Diaz R, et al. (2026). Health financing inequality in fragmented health systems: evidence from Mexico, 2000-2023. *International Journal for Equity in Health*.
- Pi S, Rutter CM, **Pineda-Antunez C**, Chen JH, Goldhaber-Fiebert JD, Alarid-Escudero F. (2026). Discrete-Event Simulation Modeling Framework for Cancer Interventions and Population Health in R (DESCIPHR): An Open-Source Pipeline. *PharmacoEconomics*. 1-19.
- Cerecero-Garcia D, Gomez-Dantes O, Hone T, **Pineda-Antunez C**, Mohar-Betancourt A, Flamand L, Servan-Mori E. (2026). Catastrophic and impoverishing health expenditures in fragmented public health systems: lessons from Mexico, 2000-2022. *Health Economics Review*.
- **Pineda-Antunez C**, Seguin C, van Duuren LA, et al. (2024). Emulator-Based Bayesian Calibration of the CISNET Colorectal Cancer Models. *Medical Decision Making*. 44(5):543-553.
- Serván-Mori E, Quezada-Sánchez AD, Fuentes-Rivera E, **Pineda-Antunez C**, Hernández-Chávez M del C, García-Martínez A, García-Feregrino R, Madrigal A, Guerrero B, Medrano G, Schnaas L (2023). Proximal determinants of suboptimal early child development during the first three years of life in socially deprived Mexican contexts. *PLoS ONE*. 18(11).
- **Pineda-Antunez C**, Seguin C, van Duuren L, Knudsen AB, Davidi B, Nascimento De Lima P, Rutter C, Kuntz KM, Lansdorp-Vogelaar I, Collier N, Ozik J, Alarid-Escudero F (2023). Emulator-based Bayesian calibration of the CISNET colorectal cancer models. *Preprint*.
- Quezada-Sánchez AD, Fuentes-Rivera E, García-Martínez A, Hernández-Chávez M del C, **Pineda-Antunez C**, Serván-Mori E (2022). Assessing the effect of an educational intervention on early childhood development among Mexican preschool children in the state of Oaxaca: a study protocol of a cluster randomized stepped-wedge trial. *Trials*. 23(1):126.
- Serván-Mori E, **Pineda-Antunez C**, Bravo-Ruiz ML, Molina M, Ramírez-Baca MI, García-Martínez A, Quezada-Sánchez AD, Orozco-Núñez E (2022). A behavioral economics analysis of the participation in early childhood development social programs promoted by civil societies in Mexico. *PLoS ONE*. 17(3):e0265389.
- Serván-Mori E, Fuentes-Rivera E, Quezada AD, **Pineda-Antunez C**, Hernández-Chávez M del C, García-Martínez A, Madrigal A, García-Feregrino R, Santiago-Angelino T, Hernández-Serrato M, Schnaas L (2022). Early neurological development and nutritional status in Mexican socially deprived contexts. *PLoS ONE*. 17(6).
- Colchero MA, Gómez R, ******Pineda-Antunez CJ**, Bautista-Arredondo S (2021). Health care utilization during the Covid-19 pandemic in Mexico: the cascade of care. *Salud Pública de México*. 63(6):743-750.
- Bautista-Arredondo S, **Pineda-Antunez C**, Cerecero-Garcia D, Cameron DB, Alexander L, Chiwevu C, Levin C (2021). Moving away from the “unit cost”: Predicting country-specific average cost curves of VMMC services accounting for variations in service delivery platforms in sub-Saharan Africa. *PLoS ONE*. 16(4).
- Diaz-Castro L, Cabello-Rangel H, **Pineda-Antunez C**, Perez de Leon A (2021). Incidence of catastrophic healthcare expenditure and its main determinants in Mexican households caring for a person with a mental disorder. *Global Mental Health*. 8.
- **Pineda-Antunez C**, Contreras-Loya D, Rodriguez-Atristain A, Opuni M, Bautista-Arredondo S (2021). Characterizing health care provider knowledge: Evidence from HIV services in Kenya, Rwanda, South Africa, and Zambia. *PLoS ONE*. 16(12).
- Siapka M, Vassall A, Cunnaman L, **Pineda-Antunez C**, Cerecero D, Sweeney S, Gomez GB (2020). Cost of tuberculosis treatment in low- and middle-income countries: Systematic review and meta-regression. *International Journal of Tuberculosis and Lung Disease*. 24(8):802-810.

- Cabello-Rangel H, Diaz-Castro L, **Pineda-Antunez C** (2020). Cost-Effectiveness Analysis of Interventions to Achieve Universal Health Coverage for Schizophrenia in Mexico. *Salud Mental*. 43(2):249-255.
- Servan-Mori E, Mendoza MA, Chivardi C, **Pineda-Antunez C**, Rodriguez-Franco R, Nigenda G (2019). A spatio-temporal cluster analysis of technical efficiency in the production of outpatient maternal health services and its structural correlates in Mexico. *International Journal of Health Planning and Management*. 34(4):e1417-e1436.
- Servan-Mori E, Cerecero-Garcia D, Heredia-Pi IB, **Pineda-Antunez C**, Sosa-Rubi SG, Nigenda G (2019). Improving the effective maternal-child health care coverage through synergies between supply and demand-side interventions: Evidence from Mexico. *Journal of Global Health*. 9(2):1-12.
- **Pineda-Antunez C**, Martinez-Silva G, Cerecero-Garcia D, Alexander L, Cameron DB, Chiwevu C, Bautista-Arredondo S (2019). Meta-analysis of average costs of HIV testing and counselling and voluntary medical male circumcision across thirteen countries. *African Journal of AIDS Research*.
- Cameron DB, Mustafa Diab M, Carroll LN, Bollinger LA, DeCormier Plosky P, **Pineda-Antunez C**, Levin C, Kahn JG (2019). The state of costing research for HIV interventions in sub-Saharan Africa. *African Journal of AIDS Research*. 18(4):277-288.
- Cerecero-Garcia D, **Pineda-Antunez C**, Alexander L, Cameron D, Martinez-Silva G, Obure CD, Bautista-Arredondo S (2019). A meta-analysis approach for estimating average unit costs for ART using pooled facility-level primary data from African countries. *African Journal of AIDS Research*.
- Rodriguez-Franco R, Servan-Mori E, Gomez-Dantes O, Contreras-Loya D, **Pineda-Antunez C** (2018). Old principles, persisting challenges: Maternal health care market alignment in Mexico in the search for UHC. *PLoS ONE*. 13(7):1-14.
- Diaz-Castro L, Cabello-Rangel H, Arredondo A, Madrigal de Leon E, **Pineda-Antunez C** (2017). Cost-effectiveness of therapeutic interventions in schizophrenia. *The European Journal of Psychiatry*. 31(1):11-16.

Talks

- 2026. Calibration for Health Decision Models via Deep Generative Models. Conference presentation, Society for Medical Decision Making Annual Meeting, Oslo, Norway.
- 2024. Quantifying the value of a potential new diagnostic test in the context of imperfect adherence. Conference presentation, Society for Medical Decision Making Annual Meeting, Boston, MA, USA.
- 2024. Fragmentation of health systems and maternal mortality transition in Mexico, 2000-2021. Conference presentation, Health Systems Global, Mexico City, Mexico.
- 2023. Optimizing a natural history simulation model of colorectal cancer using GPUs. Conference presentation, Society for Medical Decision Making Annual Meeting, Philadelphia, PA, USA.
- 2022. Emulator-based Bayesian calibration of CISNET colorectal cancer models. Poster and oral presentation, Society for Medical Decision Making Annual Meeting, Seattle, WA, USA.
- 2021. The Economic Impact of COVID-19 Non-Pharmaceutical Interventions and the influence of Poverty and Inequality. Oral presentation, International Health Economics Association, Virtual congress.
- 2020. Policy, poverty, and politics: How public health, inequality and political leadership interacted in the COVID-19 pandemic. Conference presentation, IAS COVID-19 Conference, San Francisco, CA, USA.
- 2019. Exploring the intervention costs of PMTCT and HTC: explaining and predicting unit costs for implementation scenarios using pooled data from several studies. Conference presentation, International Health Economics Association, Basel, Switzerland.
- 2018. Filling in the gaps: Analysis of VMMC unit costs using aggregated (facility-level) primary data from several studies. Conference presentation, International AIDS Conference, Amsterdam, Netherlands.

Teaching

- 2026. **Course Master, CHOICE Grad Seminar**. University of Washington, Seattle, WA, USA. Service period: March 2026-June 2026.
- 2026. **Guest lecturer, Advanced Methods in Economic and Outcomes Evaluation in Health and Medicine**. University of Washington, Seattle, WA, USA. Guest lecture delivered in February 2026.
- 2025. **Teaching assistant, Data Science in Health Economics and Outcomes Research Certificate Program**. University of Washington, Seattle, WA, USA. Service period: September 2025-June 2026.
- 2021. **Teaching assistant, Health economics, Master of Health Policy and Health Systems**. National

- Institute of Public Health, Morelos, Mexico.
- 2020. **Teaching assistant, Economic evaluation of health programs, Master of Health Economics.** National Institute of Public Health, Morelos, Mexico.
- 2019. **Teaching assistant, Workshop on Transparency and Reproducible Research.** National Institute of Public Health, Morelos, Mexico.
- 2016. **Member of the health economics faculty master's program committee.** National Institute of Public Health, Morelos, Mexico. Service period: 2016-2021.

Awards

2023, Rubenstein Endowed Memorial Scholarship, University of Washington.

2023, Graduate School Top Scholar Fellowship, University of Washington.

2023, Mexico's CONACyT Graduate Fellowship for pursuing a PhD program abroad.

2022, SMDM Lee B. Lusted Student Prize finalist in Quantitative Methods and Theoretical Developments for an outstanding presentation.

2015, Thesis honorable mention by the academic department of the INSP.

2013, Mexico's CONACyT Graduate Fellowship for Masters Studies.

2009, Acknowledge of academic excellence in undergraduate studies in biomedical engineering.

Skills

- Cost-effectiveness analysis
 - Incremental cost-effectiveness ratio
 - Value of information analysis
 - Budget impact analysis
 - Probabilistic sensitivity analysis
- Health Decision models
 - Markov models
 - Microsimulation models
 - Dynamic transmission models
 - Discrete event simulation
 - Decision trees
 - Partitioned survival models
 - Meta-modeling
- Impact evaluation
 - Causal inference
 - Quasi-experimental methods
 - Machine learning for impact evaluation
- Cost analysis
 - Microcosting
 - Cost estimation
 - Impact budgeting
- Evidence synthesis
 - Systematic reviews
 - Meta-analysis
 - Network meta-analysis
- Artificial Intelligence
 - Deep learning
 - Generative AI